

Word/Phrase	Definition & Source	Pg.
UN International Resource Panel	Launched by the United Nations Environment Programme (UNEP) in 2007 to build and share the knowledge needed to improve our use of resources worldwide. Panel consists of eminent scientists with expertise in resource management issues. Resource Panel	8
3Rs of Waste Management	Reduce, Reuse, Recycle Reduce, Reuse, Recycle: The Three R's of Waste Hierarchy (earthandhuman.org)	8
Circular Economy	Re-designing the way our economy works (e.g. designing products that can be 'made to be made again', powering the system with renewable energy... etc): Circular economy challenges our current systems through creativity and innovation so that waste can build capital. Explaining the circular economy: Rethink progress (ellenmacarthurfoundation.org)	8
Dwelling units	A single unit providing complete, independent living facilities for one or more persons, including permanent provisions for living, sleeping, eating, cooking, and sanitation. Dwelling unit Definition: 6k Samples Law Insider	8
Census Bureau	The Census Bureau's mission is to serve as the nation's leading provider of quality data about its people and economy. About the Bureau (census.gov)	8
Transit Threshold	Need more info	9
Car-sharing threshold	Need more info	9
Urban Edge	Need more info	9
Urban Core	Need more info	9
Service Economy	A service economy is an economy where the primary economic activity is the provision of services rather than the production of goods. Service Economy: Definition & Characteristics - Video & Lesson Transcript Study.com	9
Digital Economy	The digital economy is the economic activity that results from billions of everyday online connections among people, businesses, devices, data, and processes. What is digital economy? Deloitte Malta Technology	9
Upzoning, Restrictive zoning (and zoning more generally)	Upzoning means changing the zoning code to allow taller and/or denser buildings. It increases the buildable capacity of land, creating the opportunity to increase supply. Zoning more generally is the act or process of partitioning a city, town, or borough into zones reserved for different purposes (such as residence or business). What is upzoning? - Curbed	9
Cube Land Model	Chosen by the Met Council in 2010 as a platform for local real estate modeling and scenario analysis. Cube Land was chosen in part for its potential to integrate with the Council's travel demand	9

	model, allowing land use patterns and transportation network conditions to iteratively adjust over time. Forecast-Methodology-Report-(October-2019).aspx (metrocouncil.org)	
Microeconomic models	Microeconomics is concerned with the economic decisions and actions of individuals and firms. There are different theories [within microeconomic models] that emphasize certain assumption and expectations of economic behavior. Microeconomics Models and Theories - Economics Help	9
Regional Transportation Demand model	A Metropolitan Council demand forecast model, called an 'activity-based model.' Transportation decisions made by individuals ranging from long-term, day-level, and trip-level are simulated to evaluate policy and investment choices. Travel Forecasting - Metropolitan Council (metrocouncil.org) Transportation - Metropolitan Council (metrocouncil.org)	9
eGrid	EPA's Emission & Generation Resource Integrated Database Emissions & Generation Resource Integrated Database (eGRID) US EPA	10
GHG intensity	GHGs released in energy consumption for production (depends on energy sources) 04. Greenhouse gas intensity - OECD	10
AR4-IPCC Panel 4	IPCC is the Intergovernmental Panel on Climate Change, which is the United Nations body for assessing the science related to climate change. The AR4 report was released in September 2007 Fourth Assessment Report — IPCC	10
Global Warming potentials	The Global Warming Potential (GWP) was developed to allow comparison of the global warming impacts of different gases. Specifically, it is a measure of how much energy the emissions of 1 ton of a gas will absorb over a given period of time, relative to the emissions of 1 ton of carbon dioxide (CO₂). Understanding Global Warming Potentials US EPA	10
Emission Factor	An emission factor is a coefficient which allows to convert activity data into GHG emissions. It is the average emission rate of a given source, relative to units of activity or process/processes. What is an emission factor? Clim'Foot (climfoot-project.eu)	11
District Energy Systems	District energy systems are a highly efficient way to heat and cool many buildings in a given locale from a central plant. They use a network of underground pipes to pump steam, hot water, and/or chilled water to multiple buildings in an area such as a downtown district, college or hospital campus, airport, or military base. Providing heating and cooling from a central plant requires less fuel and displaces the need to install separate space heating and cooling and hot water systems in each building. Microsoft Word - district_energy_factsheet_092311 (eesi.org)	11
Powertrain	The intervening mechanism by which power is transmitted from an engine to a propeller or axle that it drives. Power Train Definition of Power Train by Merriam-Webster	12,14

Urban bus	Need more information	12
Bus rapid transit	Bus Rapid Transit (BRT) is a package of transit enhancements that adds up to a faster trip and an improved experience. METRO Red Line, A Line, and C Line. Bus Rapid Transit - Metro Transit	12
Urban rail (LRT)	Light Rail Transit is an electrically powered, two-rail technology capable of providing a broad range of passenger capacities, and operating as single vehicles or in short trains on a variety of alignment types. Rail Transit (state.mn.us)	12
Interurban rail (Northstar Commuter Rail)	Commuter rail is a passenger train service that connects an urban region together over moderate distances; which typically operates on existing freight tracks; and whose primary clientele travels between home and work. The Northstar rail line from Big Lake to Minneapolis is the first example of commuter rail in Minnesota. Rail Transit (state.mn.us)	12
2019 TBI	2019 Travel Behavior Inventory, Metropolitan Council. Household survey results on the ways in which people travel in the metro region. 2019 Travel Behavior Inventory Household Survey Results - Metropolitan Council (metrocouncil.org)	12, 16, 21,36
100-50-50 method	Defined in guide, from World Resources Institute (2012)	12
Regional Travel Demand Model	An analytical tool used for regional transportation planning from the Whatcom Council of Governments (WCOG). It is comprised of trip generation, trip distribution, mode choice, and route assignment. Regional Travel Demand Model – Whatcom Council of Governments (wcog.org)	12
FTA	Federal Transit Administration The National Transit Database (NTD) FTA (dot.gov)	13
Fuel economy	Miles per gallon (distance a vehicle can travel on one gallon of fuel) MIT School of Engineering » What's the difference between fuel efficiency and fuel economy?	13
AEO	EIA Annual Energy Outlook: an assessment by the U.S. Energy Information Administration of the outlook for energy markets through 2050. Annual Energy Outlook 2021 (eia.gov)	13,17
FHWA	Later connections include local bus (pg 14) and a freight analysis framework (pg 15)	13
Core local bus	Need more info. Simply buses operating in urban areas?	13
Suburban local bus	Need more info. Simply buses operating in suburban areas?	13
EIA Annual Energy Outlook (AEO)	Assessment made by the U.S. Energy Information Administration of the outlook for energy markets through 2050. Annual Energy Outlook 2021 (eia.gov)	14,17
Single-unit (SU) truck	“Single-unit means that a single frame supports both the power and cargo carrying portions of a vehicle. Single-unit trucks exceeding	15

	10,000 pounds maximum Gross Vehicle Weight (GVW) should be recorded here. Office of Highway Policy Information - Policy Federal Highway Administration (dot.gov)	
Combination-unit (CU) truck	Combination trucks are tractor trucks pulling trailers. Of importance is the number of axles on the truck and trailer(s), rather than the body style. Traffic Recorder Instruction Manual: Vehicle Classification Descriptions (txdot.gov)	15
Ton-miles	A statistical unit of freight transportation equivalent to a tone of freight moved one mile Ton-mile Definition of Ton-mile by Merriam-Webster	16
AADT	Annual Average Daily Traffic TFA Traffic Mapping Application - TDA - MnDOT (state.mn.us)	16
AFDC	Alternative Fuels Data Center	16,21
MA3T	Market Acceptance of Advanced Automotive Technologies: a market dynamics model that estimates the market demand for advanced vehicle powertrain technologies in response to assumptions regarding technology, infrastructure, consumers, and policies. TEEM Home (ornl.gov)	16
Long-haul freight	Need more info.	16
Embodied emissions	Includes emissions associated from the production of the car Reducing the embodied emissions of cars – The Earthbound Report	16
Per veh-km	Similar to VMT, veh-km (vkm) is a measure of traffic flow determined by multiplying number of vehicles by average length of their trips in km. OECD Glossary of Statistical Terms - Vehicle-kilometre Definition	16
Pass-km	Need more info	16
GREET model	The Greenhouse gases Regulated Emissions and Energy in Transportation model; a publicly available spreadsheet model developed at the Argonne National Laboratory (ANL) that estimates emissions from CO ₂ , CH ₄ , and N ₂ O and criteria pollutants from transportation fuels used in the United States. GREET Model - Life Cycle Associates, LLC	17
MRO	Midwest Reliability Organization Meetings (midwestreliability.org)	17
Macroeconomic conditions	A macroeconomic factor is a pattern, characteristic, or condition that emanates from, or relates to, a larger aspect of an economy rather than to a particular population. The characteristic may be a significant economic, environmental, or geopolitical event that widely influences a regional or national economy. Macroeconomic Factor - Overview, Components, Types (corporatefinanceinstitute.com)	17
Brent crude oil	Brent is a marker grade of oil, with specific physical characteristics (light, low sulfur content) used as a guideline in determining the quality of oil.	17

	What Is Brent Crude Oil Monfex	
NREL FASTim tool	The Future Automotive Systems Technology Simulator (FASTim), provides a simple way to compare powertrains and estimate the impact of technology improvements on light-, medium-, and heavy-duty vehicle efficiency, performance, cost, and battery life. FASTSim: Future Automotive Systems Technology Simulator Transportation and Mobility Research NREL	18
Revenue-miles	A transportation metric that shows the number of miles traveled by paying passengers. Typically used as an airline traffic statistic. revenue-miles - Search (bing.com)	19
S-curve technology adoption projection	Describes how growth processes (limited stocks) cannot be mathematically by a constant growth or linear function; it shows an initial exponential growth until an inflection point of exponential decay from then on until reaching the upper asymptote (exhausting the stock) The S-curve as a Forecasting Tool (areppim.com)	19
ZCAP	America's Zero Carbon Action Plan (ZCAP), developed by SDSN USA; a network of universities and research institutions across the United States committed to sustainable development SDSN USA	21
Low-carbon fuel standard (LCFS)	Approved in 2009, implemented in 2011 in California. The LCFS is designed to encourage the use of cleaner low-carbon transportation fuels in California, encourage the production of those fuels, and therefore, reduce GHG emissions and decrease petroleum dependence in the transportation sector. Low Carbon Fuel Standard California Air Resources Board	22
Marginal generation	Certain types of energy producing sources (e.g. coal and natural gas) are disproportionately used as a marginal resource (generating resource that responds to changes in demand). A source like nuclear power, on the other hand, has very limited capacity to adjust output and is producing most of the time at maximum capacity. Why a 'marginal' source is central to climate change (taf.ca)	23
5 th IPCC assessment	IPCC is the Intergovernmental Panel on Climate Change, which is the United Nations body for assessing the science related to climate change. The AR5 (5 th IPCC assessment) report was released in October 2014 Fifth Assessment Report — IPCC	23
tCO ₂ e	Tons of CO ₂ equivalent. A metric used to compare the emissions from various greenhouse gases on the basis of their global-warming potential (GWP), by converting amounts of other gases to the equivalent amount of carbon dioxide with the same global warming potential. Glossary:Carbon dioxide equivalent - Statistics Explained (europa.eu)	23

Price parity	Need more info regarding EVs. When the price difference between something like electric vehicles and internal combustion vehicles becomes 0. Smaller electric vehicles to reach price parity with ICE counterparts by 2028 AFTmedia (autofintechs.com)	24
BNEF	BloombergNEF; a team of experts across six continents assessing pathways for the power, transport, industry, buildings and agriculture sectors to adapt to the energy transition. They help commodity trading, corporate strategy, finance and policy professional navigate change and generate opportunities. About Us BloombergNEF (bnef.com)	24
Cross elasticities	Need more info. Cross elasticity of demand is an economic concept that measures the responsiveness in the quantity demanded of one good when the price for another good changes. In the context of transit ridership, estimating how it is affected by a variety of automobile policy strategies. Cross Elasticity of Demand Definition & Formula (investopedia.com)	26
Network Next	A 20-year plan from metro transit for expanding and improving bus network. Transit improvements under consideration include improved local and express routes, integrated shared mobility options, and new arterial bus rapid transit (BRT) lines. Network Next - Metro Transit	26
Off-board fare collection	Bus riders pay with a MetroCard or coins at a sidewalk ticket vending machine at the Select Bus Service bus stop before they get on the bus. Riders may then board the bus through any door, without needing to show the ticket to the bus operator. Bus Rapid Transit - Select Bus Service Features (nyc.gov)	26
Marginal abatement cost	The marginal cost of abatement refers to the marginal loss in company profits from avoiding the last unit of pollution or the marginal cost of reaching a specified pollution target because of some degree of output. Marginal Cost of Abatement: Understanding the Basics (sinaitechnologies.com)	26
LCCA	Life Cycle Cost Analysis?	26
Overhead catenary line	An electric cable used to transmit electrical energy to electric locomotives, trolleybuses, or trams. Overhead line - Wikipedia	26
Elasticities	Need more info – will help to have a project-specific definition.	27
Meta-analysis	“Meta-analysis refers to the statistical analysis of the data from independent primary studies focused on the same question, which aims to generate a quantitative estimate of the studied phenomenon, for example, the effectiveness of the intervention” Frontiers Meta-Analytic Methodology for Basic Research: A Practical Guide Physiology (frontiersin.org)	28
CBD	Need more info. Central Business District?	29, 38

Land use entropy measure/entropy term	Need more info. Entropy is defined as a measure of randomness or disorder of a system – the number of ways in which a system can be arranged. In the context of forecasted land uses, the entropy term appears to measure the amount of diverse land use types over time.	30
OpenStreetMap	OpenStreetMap provides map data for thousands of web sites, mobile apps, and hardware devices. It's built by a community of mappers that contribute and maintain data about roads, trails, cafes, railway stations, and much more, all over the world. OpenStreetMap	30
GTFS	General Transit Feed Specification (GTFS) is a data specification that allows public transit agencies to publish their transit data in a format that can be consumed by a wide variety of software applications. General Transit Feed Specification (gtfs.org)	31
MAPC	The Metropolitan Area Planning Council (MAPC) of Massachusetts, is the regional planning agency serving the people who live and work in the 101 cities and towns of Metropolitan Boston. About us – MAPC	31, 37
VisionEval transportation model	VisionEval is a collaborative project to build a family of strategic tools for performance-based transportation planning into a single open-source programming framework. Strategic tools are designed to evaluate many alternative futures and policies to help state and metropolitan area governments address pressing issues, despite uncertainty. VisionEval · VisionEval	31,38
UrbanSim land use model	UrbanSim is a simulation system for supporting planning and analysis of urban development, incorporating the interactions between land use, transportation, the economy and the environment. It is designed for use by Metropolitan Planning Organizations (MPOs), cities, counties, non-governmental organizations, real estate professionals, planners, researchers and students interested in exploring the effects of infrastructure and development constraints as well as other policies on community outcomes such as motorized and non-motorized accessibility, housing affordability, greenhouse gas emissions, and the protection of open space and environmentally sensitive habitats. UrbanSim — UrbanSim	31,38
Restrictive zoning	A zoning restriction dictates the way land in a jurisdiction can be used. For example certain vacant lots will be 'zoned' for either residential or commercial purposes. Zoning Restrictions VS. Deed Restrictions - LandCentral	
Driving cycle efficiency	Driving cycles consisting of unique profile of stopping, starting, idling, constant speed cruising, acceleration, and deceleration with a prescribed time frame, vary considerably worldwide, depending on specific traffic and road conditions. A driving cycle is a set of values (generally velocity vs time), which represents how a vehicle travels	32,34

	through a path. Efficiency of the driving cycle helps determine the performance and energy consumption of vehicles. Driving Cycle - an overview ScienceDirect Topics What is a Driving Cycle and How to Develop One for EV simulation? (getelectricvehicle.com)	
Platooning	Vehicle platooning is part of a suite of features that self-driving cars might employ. A platoon is a group of vehicles that can travel very closely together, safely at high speed. What is Vehicle Platooning? (drivingtests.co.nz)	33
Bass diffusion model	The Bass diffusion model, is a differential equation used to describe the process involved in the adoption of new products into the population. It is used to present a rationale of the interaction between current and potential adopters of a new product. Bass Diffusion Model - Explained - The Business Professor, LLC	33,35
Wrt	Need more info	34
The Minnesota CAV Challenge	The Minnesota CAV Challenge is a new, innovative request for proposals that seeks to advance emerging technologies to improve transportation for all Minnesotans. Connected and Automated Vehicle (CAV) technology is advancing rapidly. CAV Challenge - Connected and Automated Vehicles - MnDOT (state.mn.us)	35
Ride-hail travel	Ride hail is a service provided by companies such as Lyft and Uber Ride hail OHSU	36
Micro-mobility trip distance	Micro-mobility is a term used to describe travel solutions for short distances, usually the first or last mile of a journey. Solutions include light weight vehicles that are usually electric, like scooters bikes, and other modes of transportation. What is micromobility? Facts about small electric vehicles in cities SEAT	36
TAM team/study	Transit Asset Management (TAM) rule of the Federal Transit Administration (FTA) requires transit agencies, metropolitan planning organizations (MPO), and state agencies to work together to achieve and maintain a state of good repair for transit systems. CTPS Technical Memo	38
Non-transpondered vehicles	A vehicle without a device for receiving a radio signal and automatically transmitting a different signal. In the context of road pricing, vehicles without GPS transponders would be harder to track and regulate, which is why the authors state that charging a higher fixed rate for non-transpondered vehicles could help speed up the rate of adoption.	39
Parkade	Multistory parking garage	39
Pre-loaded smart cards	Smart cards are often used as transit passes. When riders purchase these cards, they are given a certain number of credits to use on their ride. In the context of behavioral nudges, a sunk cost into smart cards for public transportation could encourage more frequent transit use. SmartCards.net – Smart Cards and Microprocessors	43

